00-S-305C
January 10, 1972
SUPERSEDING
Fed. Spec. 00-S-305B
March 7, 1967

FEDERAL SPECIFICATION

SHEARING MACHINE, METAL SQUARING, MECHANICAL DRIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mechanically driven, metal squaring shears.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. The machines covered by this specification shall be of the types, classes, styles, and sizes, as specified (see 6.2.1). The machine types shall be as listed below. The sizes of the type I machines shall be as listed in table I. The classes, styles, and sizes of the type II machines shall be as listed in table II.

Type I - Gapless or gap frame (see table I).
Type II - Gap frame (see table II).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 595 - Color.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices included in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC , Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-M-18058 - Machinery, Metal, and Woodworking Support Equipment and Associated Repair Parts, Preparation for Delivery of.

Military Standards:

MIL-STD-129 - Marking for Shipping and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI), Inc., Standards:

B6.1 - Tooth Proportions for Coarse-Pitch Involute Spur Gears.
B6.7 - Tooth Proportions for Fine-Pitch Involute Spur and Helical Gears.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

JOINT INDUSTRIAL COUNCIL (JIC):

Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the National Machine Tool Builders' Association, 2139 Wisconsin Avenue, N.W., Washington, DC 20007.)

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. HANDBOOK:

SAE J403D - SAE Standard for Chemical Compositions of SAE Carbon Steel.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1 and 6.3), the supplier shall furnish one complete machine of each type, class, style, and size specified in the contract or order for first article inspection and approval in accordance with section 4 herein.

3.2 Design. The shearing machine shall be a self-contained, electrically powered, mechanically driven unit of the manufacturer's current design. The machine shall be designed to accomplish shearing by the downward movement of a ram mounted, upper shear knife past a table or machine

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bed mounted, stationary and lower shear knife. All components subject to frequent adjustment or replacement shall be easily accessible.

3.2.1 Safety devices. The shearing machine shall be furnished with safety devices or coverings for rotating shafts, flywheels, gears, pulleys, belts, and all other components that present safety hazards. The devices shall include an upper shear blade finger guard for protecting the operator during the shearing operation. The devices shall provide easy access to the guarded components. When special safety devices are required, they shall be as specified (see 6.2.1).

3.2.1.1 Department of the Navy. The shearing machine procured for shipboard application by the Department of the Navy shall be provided with safety devices to prevent accidental tripping of the operating controls of the drive (see 6.2.1.b).

3.2.2 Lubrication. Bearings (except sealed, permanently-lubricated type), mating gears, rotating, and sliding components shall be provided with means for lubrication. The ram guide bearings and the bearings and gears of the drive shall be provided with a positive means of lubrication. Unless otherwise specified (see 6.2.1), a positive means of lubrication shall be provided by either a one-shot type or automatic type lubrication system for size # 10 gage and smaller capacity machines, and an automatic type lubrication system shall be provided for size 3/16 inch and larger capacity machines. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.2.3 Interchangeability. All components shall be manufactured to such standards as will permit replacement or adjustment without modification of components or machine.

3.2.4 Material. All material used in the manufacture of the machine shall be new and of the quality necessary to produce a machine to meet the requirements described herein.

3.2.5 Gears. All gears shall meet the applicable requirements of ANSI B6.1 and ANSI B6.7.

3.3 Construction. All components used in the construction of the machine shall be new and sound in structure.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding,

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peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are required in the original design. These operations shall not be employed as repair measures for defective components.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded components, stampings, and welded parts shall be clean and free from harmful or extraneous materials such as sand, dirt, fins, sprues, scale, and flux. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the shearing machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.5.1 Department of the Navy. The shearing machine procured for shipboard application by the Department of the Navy shall be painted light gray conforming to color no. 16187 of Fed. Std. No. 595 (see 6.2.1.b).

3.3.6 Threads. All threaded components shall be in the inch system and shall conform to H28.

3.4 Performance. The machine shall be capable of shearing SAE 1020 steel per J403D to comply with the workpiece size and shearing speed capacities specified in table I or II as applicable for the machine type, class, style, and size specified in the contract or order. The resulting sheared edge of the table cut portion of the workpiece shall be within ± 1-1/2 angular degree of perpendicular (90°) with the flat side of the workpiece. The resulting sheared edge of the off-cut portion of the workpiece shall be within ± 3 angular degrees of perpendicular (90°) with the flat side of the workpiece. The type II machine shall also be capable of slitting metal to obtain workpiece off-cut widths of sizes equal to the gap depth dimension

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specified in table II as applicable for the machine class specified. The sheared workpieces shall be free of holddown system damage.

3.4.1 Gage shearing. The workpiece positioning gages shall be capable of utilization with the machine for shearing workpieces of any size up to and including the width capacity and not less than 1/2 the thickness capacity of the machine, to precision tolerances as follows:

a. Parallel shear - within ± 0.0015 inch per linear foot of workpiece width, parallel with a sheared (trimmed) edge positioned to the back gage.

b. Square shear - within ± 0.0015 per linear foot of workpiece width, square with a sheared (trimmed) edge positioned to either side squaring gage.

c. Dimension shear - within ± 0.005 inch of the back gage setting.

3.5 Size, capacities, and dimensions. Unless otherwise specified (see 6.2.1), the size, capacities, and dimensions of the machine shall be as specified in table I or II for the applicable machine type, class, style, and size specified in the contract or order.

3.6 Components. The machine shall be provided with components to include a frame, ram, drive, holddown system, gages, and shear knives.

3.6.1 Frame. The frame shall include two cast iron or steel plate end housings with connecting table and cross members, joined together to form a single unit for supporting all other components of the machine. The frame members shall be joined together by interlocked and bolted construction to permit assembly and disassembly. The end housings of the type II machine shall be provided with gaps or workpiece openings to comply with table II. The base of the end housings shall have provisions for securing the machine to a flat mounting surface.

3.6.1.1 Table. The workpiece table or bed shall be constructed of either cast iron or steel plate, with the top surface finished to 125 microinches or less, and flat to a tolerance of ± 0.005 inch per linear foot. The off-cut edge of the table shall be provided with means for securing and supporting the lower shear knife in the stationary shearing position. Either the table or ram shall be provided with an adjustment for obtaining and maintaining clearance alignment between the upper and

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lower shear knives. The top of the table shall be provided with hand slots for facilitating the operator in handling the workpieces. The top of the table shall also be provided with holes, slots, or keyways, as required, for attaching and accommodating workpiece positioning gages specified herein.

3.6.2 Ram. The ram shall be constructed of either cast iron or steel plate, designed with sufficient size and rigidity for securing and supporting the upper shear knife in the shearing action. The stroke of the ram shall be accomplished on guide bearings of a design required for assuring shearing accuracy specified herein. A knife seat shall be machined across the full width of the ram for directly supporting the upper shear knife. The upper shear knife shall be bolted to the ram in an inclined position to comply with the rake angle requirements of tables I or II, as applicable for the type, class, style, and size shear specified in the contract or order. Unless otherwise specified (see 6.2.1), the ram for the type II, size 1/4 inch and larger capacity machines shall be vertically adjustable for performing metal slitting, notching, and square shearing. The adjustment shall permit the ram to be adjusted to a position necessary for accomplishing the shear stroke without the high end of the upper shear knife passing the shear line of the lower shear knife.

3.6.3 Drive. The machine shall be provided with an electric motor driven mechanical type drive, designed to develop and control the reciprocating stroke of the ram. The drive motor shall be supplied supplementary power by a flywheel of cast iron or steel, and dynamically balanced after machining. An eccentric shaft shall be provided for converting the rotary motion of the motor and flywheel to the reciprocating motion required by the ram for accomplishing shearing performance specified herein. The eccentric shaft shall rotate within bronze bushings or anti-friction bearings and be directly supported by the end housings of the frame. The energy of the drive shall be controlled by a clutch arrangement capable of transmitting the maximum developed torque of the drive to the eccentric shaft and ram. The clutch shall be provided with a brake for stopping the motion of the ram at the top of the ram stroke when the control is set for single stroke operation.

3.6.3.1 Drive operating controls. The drive shall be provided with controls for operating the ram to provide both single (non-repeat) and continuous stroke operation. Unless otherwise specified (see 6.2.1), the drive shall be operated by either a treadle type or foot switch type control.

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3.6.4 Holddown system. The machine shall be provided with a holddown system for securing to the table, any workpiece size up to and including the capacity of the machine. The holddown system shall include hydraulic, pneumatic, or spring operated type plunger-cylinder units spaced not more than 12 inches between centers across the full workpiece opening of the machine. The plunger-cylinder units shall be arranged to apply uniform holddown pressure, preventing workpiece movement that would impede the specified shearing performance. Operation of the plunger-cylinder units shall be automatically controlled by the machine to secure the workpiece to the table on the upper shear knife downstroke, and release the workpiece on the upstroke. When specified (see 6.2.1), hydraulic type holddown systems shall also be operational by a manual control that functions independently of the shear stroke to lower and raise the plunger-cylinder units. The manual control shall be provided with means for controlling holddown pressure of the plunger-cylinder units.

3.6.5 Gages. The machine shall be provided with workpiece positioning gages to include a back gage, front gage, and side squaring gage. The gages shall accommodate any workpiece size ranging up to and including the capacity of the machine.

3.6.5.1 Back gage. A back gage shall be furnished on the rear of the machine to provide an adjustable reference edge for positioning the workpiece parallel with the stationary shear knife. The reference edge shall be adjustable for any position within the back gage capacity specified herein. Unless otherwise specified (see 6.2.1), the size # 10 gage and smaller capacity machines shall be furnished with either a power or manually operated type back gage; the size 3/16 inch and larger capacity machines shall be furnished with a power operated type back gage. The operating controls of the power operated type back gage shall be located on the front of the machine. The operating controls of the power operated type back gage shall be provided with a direct reading indicating dial type position indicator for indicating the distance between the reference edge and the shear line of the stationary shear knife. When specified (see 6.2.1), the manually operated type back gage shall be provided with a direct reading indicating dial type position indicator with operating controls located on the front of the machine. When specified (see 6.2.1), the back gage shall be furnished with a magnetic workpiece support mechanism for supporting the off-cut portions of capacity width workpiece sizes of # 16 gage and less. Adjustment shall be provided for correcting misalignment of the back gage.

3.6.5.2 Front gage. At least two detachable type, front gage support arms, with adjustable workstops, shall be provided on the front of the table for positioning the workpiece parallel to the stationary shear knife. The

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workstops shall be adjustable in T-slots or grooves, machined throughout the full length of the support arms. Means shall be provided for locking the workstops in position.

3.6.5.3 Side squaring gage. Both the right and left hand sides of the table shall be provided with a detachable type side squaring gage for positioning the workpiece at right angles to the stationary shear knife for shearing. Graduated measuring scales shall be recessed in either the side squaring gages or the table top for measuring the workpiece. When required, a side squaring extension arm shall be furnished with the machine, with length as specified (see 6.2.1). The side squaring gage extension arm shall be provided with a graduated scale and adjustable workstop.

3.6.5.4 Slitting gage. When specified (see 6.2.1), the type II machine shall be provided with a detachable type slitting gage for guiding the workpiece to accomplish slitting.

3.6.5.5 Light beam or light mirror type shearing gage. When specified (see 6.2.1), the machine shall be provided with either a light beam or light mirror type shearing gage that utilizes light beam shadows at the shear line for shearing to a line scribed on the workpiece. The light beam or light mirror type shearing gage shall be wired through the electrical controls of the machine.

3.6.6 Shear knives. Unless otherwise specified (see 6.2.1), both the ram and table shall be provided with shear knives of tool steel, hardened to at least 52 Rockwell C, and machine finished for endurance and accuracy. The shear knives shall be the reversible type with four cutting edges.

3.6.6.1 Department of the Navy. The shearing machine procured for shipboard application by the Department of the Navy shall be furnished with an additional set of shear knives (see 6.2.1.b).

3.7 Electrical components. Unless otherwise specified (see 6.2.1), all electrical components shall conform to, and be located on, the machine in accordance with JIC Electrical Standards for General Purpose Machine Tools.

3.7.1 Motor. Unless otherwise specified (see 6.2.1), the machine drive motor shall be 230/460 volt, 3 phase 60 Hz, continuous duty, squirrel-cage induction type, and wired for operation on the voltage specified (see 6.2.1). The motor shall be provided with sealed and permanently lubricated type bearings.

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3.7.2 Electrical controls. The machine shall be completely wired through the controls and ready for connection to the applicable power source. The machine drive motor control shall be equipped with a magnetic starter having overload and low voltage protection with a manual reset. Control circuit voltage shall be nominal 115 volts, single phase, obtained from a transformer with an isolated secondary winding. The machine shall be provided with operating controls to comply with 3.6.3.1.

3.8 Equipment.

3.8.1 Standard equipment. Unless otherwise specified (see 6.2.1), one set of wrenches or tools, normally furnished with the machine by the manufacturer shall be furnished.

3.8.2 Optional equipment. When specified (see 6.2.1), the following optional equipment or features shall be furnished with each machine:

a. Vernier or direct dial indicating type pin gage arrangement.

b. Other optional equipment or features as required.

3.8.3 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.9 Electromagnetic interference control. When specified (see 6.2.1), the machine shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, marked in accordance with MIL-STD-130, shall be securely affixed to the machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature.
Manufacturer's name.
Manufacturer's model designation.
Manufacturer's serial number.
Power input characteristics and ratings.

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Date of manufacture.
Contract or order number.
Federal Stock Number or Plant Equipment Code.
U.S.

3.11 Identification plate. When specified (see 6.2.1), a separate identification plate of corrosion resistant metal, permanently and legibly marked in letters 3/16 inches high with the caption "Departmental or Activity Identification" shall be securely affixed to each machine. The plate shall have blank space for the Government to stamp 25 letters or numbers of the same size as the caption.

3.12 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be securely affixed to each machine. The following information shall be furnished on the plate:

Type of lubricant.
Viscosity.
Federal or Military Specification Number.

3.13 Workmanship. Workmanship of the machine and accessories shall be of the quality prevailing among manufacturers normally producing equipment of the type specified herein.

3.14 Fungus control. When required, fungus proofing shall be as specified (see 6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or one or more of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination or any test shall be cause for rejection.

4.5 Examination. The machine and associated equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load for not less than 30 minutes, during which time the drive, holddown system, and gages shall be actuated and observed to determine compliance with the requirements specified herein.

4.6.2 Performance test. Machine performance shall be tested by operating the machine to shear SAE 1020 steel, per SAE J403D. The instruments and methods used in performing the test shall be at the option of the manufacturer provided they determine the capacity and gage shearing capability of the machine.

4.6.2.1 Capacity shearing test. Capacity shearing shall be verified by performing shear cuts in the workpiece size and the shearing speed capacities specified in table I or II as applicable for the type, class, style, and size machine specified in the contract or order. The shear cuts shall be performed to comply with 3.4.

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4.6.2.2 Gage shearing test. Gage shearing shall be verified by utilizing the back gage and the side squaring gage to perform parallel and square shearing in workpiece sizes of not less than the full width capacity, and not less than 1/2 the thickness capacity, specified in table I or II as applicable for the type and size machine specified. Gage shearing shall be performed to comply with 3.4.1.

4.6.3 Electromagnetic interference control test. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control shall be tested in accordance with the applicable requirements of MIL-STD-461 to determine conformance with 3.9.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2.1), preservation, packaging, and packing shall be in accordance with the requirements of MIL-M-18058. The levels of preservation, packaging, and packing shall be as specified (see 6.2.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required in the contract or purchase order, interior packages, and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Unless otherwise specified (see 6.2.1), marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. The shearing machines covered by this specification are intended for use in metal squaring operations that are generally required in all phases of metal fabricating work.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

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6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Specify if Department of the Navy procurement is for shipboard application.

c. Type, class, style, and size machine required (see 1.2.1 and tables I and II).

d. First article, if required (see 3.1).

e. Special safety devices, if required (see 3.2.1).

f. Lubrication, if different (see 3.2.2).

g. Painting, if different (see 3.3.5).

h. Size, capacities, and dimensions of the machine, if different (see 3.5).

i. Vertically adjustable ram, if different (see 3.6.2).

j. Drive operating controls, if different (see 3.6.3.1).

k. Manually operated holddown control for hydraulic systems, if required (see 3.6.4).

l. Back gage operation, if different (see 3.6.5.1).

m. Direct reading indicating dial style position indicator for manually operated type back gage, if required (see 3.6.5.1).

n. Magnetic workpiece support mechanism for back gage, if required (see 3.6.5.1).

o. Side squaring extension arm, if required, specify length desired (see 3.6.5.3).

p. Slitting gage, if required (see 3.6.5.4).

q. Light beam or light mirror shearing gage, if required (see 3.6.5.5).

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r. Shear knives, if different (see 3.6.6).

s. Electrical components, if different (see 3.7).

t. Motor characteristics, if different (see 3.7.1).

u. Voltage for initial wiring of drive motor (see 3.7.1).

v. Standard equipment, if different (see 3.8.1).

w. Optional equipment, if required, specify equipment or features (see 3.8.2).

x. Repair parts, if required (see 3.8.3).

y. Electromagnetic interference control, if required (see 3.9).

z. Nameplate data, if different (see 3.10).

aa. Identification plate, if required (see 3.11).

bb. Lubrication plates, if different (see 3.12).

cc. Fungus control, if required (see 3.14).

dd. First article inspection, if different (see 4.3).

ee. Quality conformance inspection, if different (see 4.4).

ff. Electromagnetic interference control test, if different, specify tests required (see 4.6.3).

gg. Applicable levels of preservation, packaging, and packing (see 5.1).

hh. Marking, if different (see 5.2.2).

6.2.2 Contract data requirements. In military procurements, technical data such as the manufacturer's standard operators manual, parts list, foundation and anchor bolt plans, and wiring diagrams will usually be required for delivery in connection with this specification. When technical data is required, it will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

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6.3 Instructions for first article approval. When first article inspection and approval are required for equipment furnished under this specification for Military activities, the solicitation shall conform to the applicable requirements of Section I, Part 19, of the Armed Services Procurement Regulation.

6.4 Cross reference of classification. A cross reference of machine classifications between 00-S-305B and 00-S-305C are as follows:

00-S-305B	00-S-305C
Type I - Gapless frame.	Type I - Gapless or gap frame.
Type II - Gap frame.	Type II - Gap frame.
Class A - Extra shallow gap, 1" Min.	Type I - Gapless or gap frame.
Class B - Shallow gap, 3" Min.	Type I - Gapless or gap frame.
Style 1 - Slow speed (deleted).	
Style 2 - Medium speed (deleted).	
	Type II - Gap frame.
	Class A - 6" Min. gap (New).
	Style 1 - Low rake angle (New).
	Style 2 - High rake angle (New).
Class C - Medium gap, 18" Min.	Class B - 18" Min. gap.
Style 1 - Slow speed (deleted).	
Style 2 - Medium speed (deleted).	
	Style 1 - Low rake angle (New).
	Style 2 - High rake angle (New).
Class D - Deep gap, 24" Min.	Class C - 24" Min. gap.
	Style 1 - Low rake angle (New).
	Style 2 - High rake angle (New).
Class E - Extra deep gap, 36" Min.	Class D - 36" Min. gap.
	Style 1 - Low rake angle (New).
	Style 2 - High rake angle (New).

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TABLE I.

Type I. Gapless or Gap Frame Machine (Manufacturer's Option)

Machine Size, Shearing Capacity, (Inches) Not Less Than		Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than	Back Gage, Capacity, (Inches) Not Less Than	Front Gage Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics (Inches) Not More Than		
Workpiece Thickness	Workpiece Width					Width	Depth (w/o Front Gage)	Height
#16 Gage (0.0598)	36	80	3/8	24	32	69	40	70
	42	80	3/8	24	32	75	40	70
	48	80	3/8	24	32	85	40	70
	60	80	3/8	24	32	95	40	70
	72	60	5/16	24	32	108	40	70
	96	60	5/16	24	32	142	59	74
	120	60	5/16	24	32	168	59	74
	144	60	5/16	24	32	198	64	74
	216	60	5/16	24	32	282	104	74
#14 Gage (0.0747)	36	60	13/32	24	32	76	40	70
	42	60	13/32	24	32	76	40	70
	48	60	13/32	24	32	92	60	70
	60	60	13/32	24	32	98	60	70
	72	60	13/32	24	32	118	60	70
	96	60	3/8	24	36	160	66	60
	120	60	3/8	24	36	160	66	60
	144	60	3/8	24	36	198	66	60
#12 Gage (0.1046)	24	60	11/32	24	32	59	40	52
	36	60	11/32	24	36	88	40	52
	48	60	3/8	24	36	98	77	70
	60	60	3/8	24	36	107	77	70
	72	60	3/8	24	36	123	77	70
	96	60	3/8	24	36	152	77	74
	120	60	3/8	24	36	176	77	74
	144	60	3/8	24	36	198	77	74
#10 Gage (0.1345)	24	60	7/16	24	32	74	54	56
	36	60	5/16	24	32	76	54	60
	48	60	7/16	24	32	102	77	79
	60	60	7/16	24	36	112	77	79
	72	60	3/8	24	36	126	77	79
	96	60	3/8	24	36	160	77	79
	120	60	3/8	24	41	200	77	79
	144	60	3/8	24	41	224	77	79
	168	60	1/4	24	41	236	77	79
	192	60	1/4	24	41	268	77	79

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TABLE I.(Cont'd)

Type I. Gapless or Gap Frame Machine (Manufacturer's Option)

Machine Size, Shearing Capacity, (Inches) Not Less Than		Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than	Back Gage, Capacity, (Inches) Not Less Than	Front Gage, Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics (Inches) Not More Than		
Workpiece Thickness	Workpiece Width					Width	Depth (w/o Front Gage)	Height
3/16"	24	60	7/16	24	36	74	54	56
	36	60	7/16	24	36	92	68	66
	48	60	3/8	24	36	128	84	80
	72	60	3/8	24	36	152	84	80
	96	60	3/8	24	36	176	84	80
	120	60	3/8	24	41	198	84	80
	144	55	3/8	24	41	226	84	80
	168	55	5/16	24	41	249	88	80
	192	50	5/16	24	41	299	90	88
	216	40	5/16	24	41	326	90	88
1/4"	48	60	3/8	24	41	128	76	80
	72	60	3/8	24	41	152	76	80
	96	60	3/8	36	41	176	88	80
	120	60	3/8	36	41	198	86	80
	144	60	3/8	36	41	226	86	80
	168	50	5/16	36	45	249	88	86
	216	30	5/16	36	45	299	90	86
	240	25	5/16	36	45	326	106	96
3/8"	48	60	1/2	36	48	124	84	80
	72	50	1/2	36	48	169	84	86
	96	50	3/8	36	48	195	88	86
	120	50	5/16	36	48	220	88	86
	144	50	5/16	36	48	246	88	86
	168	40	23/64	36	48	248	90	86
	192	40	21/64	36	48	276	90	86
	216	30	5/16	36	56	328	106	96
	240	30	5/16	36	56	349	106	96
1/2"	48	50	1/2	36	48	134	92	86
	72	50	1/2	36	48	170	92	86
	96	40	13/32	36	48	196	90	84
	120	40	13/32	36	48	220	90	84
	144	40	13/32	36	48	242	90	84

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TABLE I.(Cont'd)

Type I. Gapless or Gap Frame Machine (Manufacturer's Option)

Machine Size, Shearing Capacity, (Inches) Not Less Than		Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than	Back Gage, Capacity, (Inches) Not Less Than	Front Gage, Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics (Inches) Not More Than		
Workpiece Thickness	Workpiece Width					Width	Depth (w/o Front Gage)	Height
5/8"	48	40	1/2	36	48	128	90	84
	72	40	1/2	36	48	149	90	84
	168	20	1/2	36	48	278	115	102
	192	20	1/2	36	48	298	115	102
3/4"	96	30	1/2	36	48	204	100	92
	120	30	1/2	36	48	228	100	92
	144	30	1/2	36	48	258	100	92
1"	48	30	5/8	36	48	156	100	92
	72	30	5/8	36	48	179	100	92

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TABLE II.

Type IL Gap Frame Machines

Class A - * Gap Depth Not Less Than 6 Inches
Class B - * Gap Depth Not Less Than 18 Inches
Class C - * Gap Depth Not Less Than 24 Inches
Class D - * Gap Depth Not Less Than 36 Inches

Style 1 - Low Rake Angle
Style 2 - High Rake Angle

Machine Class	Machine Size Shearing Capacity, (Inches) Not Less Than		Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than		Back Gage, Capacity, (Inches) Not Less Than	Front Gage Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics, (Inches) Not More Than		
	Workpiece Thickness	Workpiece Width		Style 1	Style 2			Width	Depth (w/o Front Ga.)	Height
B	#14 Gage (o.0747)	36	60	7/16	**	24	36	80	60	65
B		48	60	7/16	**	24	36	96	60	65
B		72	60	7/16	**	24	36	118	60	65
B		96	60	7/16	**	24	40	150	60	84
B		120	60	7/16	**	24	40	170	60	84
A	#12 Gage (o.1046)	48	65	1/4	**	36	42	86	80	68
A		72	65	1/4	**	36	42	112	80	68
A		96	65	1/4	**	36	42	136	80	68
A		120	65	1/4	**	36	42	160	80	68
A		144	65	1/4	**	36	45	184	80	68
A		156	65	1/4	**	36	45	196	80	72
A		168	65	1/4	**	36	45	210	80	72
A	#10 Gage (o.1345)	48	65	1/4	**	36	42	86	80	68
A		72	65	1/4	**	36	42	112	80	68
A		96	65	1/4	**	36	42	136	80	68
A		120	65	1/4	**	36	45	160	80	68
A		144	65	1/4	**	36	47	184	80	68
A,B		168	60	1/4	**	36	47	210	80	80
B		192	50	1/4	**	36	47	236	86	88
A	3/16"	48	65	3/8	**	36	42	86	80	68
B		48	60	1/4	**	36	42	86	80	68
A		72	65	3/8	**	36	42	112	80	80
B		72	60	1/4	**	36	42	112	80	80
A		96	65	3/8	**	36	42	136	80	80
B		120	60	1/4	**	36	45	160	80	80
B		144	60	1/4	**	36	47	184	80	80
B		156	60	1/4	**	36	47	198	80	80
B		168	60	1/4	**	36	47	208	80	80
B		192	50	1/4	**	36	50	238	84	94
B		216	40	1/4	**	36	50	260	88	94

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TABLE II. (Cont'd)

Type II. Gap Frame Machines

Class A - * Gap Depth Not Less Than 6 Inches
Class B - * Gap Depth Not Less Than 18 Inches
Class C - * Gap Depth Not Less Than 24 Inches
Class D - * Gap Depth Not Less Than 36 Inches

Style 1 - Low Rake Angle
Style 2 - High Rake Angle

Machine Class	Machine Size Shearing Capacity, (Inches) Not Less Than: Workpiece Thickness	Workpiece Width	Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than: Style 1	Style 2	Back Gage, Capacity, (Inches) Not Less Than	Front Gage Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics, (Inches) Not More Than: Width	Depth (w/o Front Ga.)	Height
B	1/4"	48	60	1/4	5/16	36	45	90	80	80
B		72	60	1/4	5/16	36	45	112	80	80
B		96	60	1/4	5/16	36	45	136	80	80
B		120	60	1/4	5/16	36	47	160	80	86
B		144	60	1/4	5/16	36	47	184	80	86
B		168	50	1/4	**	36	50	215	86	86
B	1/4"	192	40	1/4	**	36	50	250	90	94
B		216	30	5/16	**	36	50	274	100	100
B		240	25	7/32	**	36	50	310	106	102
C		264	25	3/16	**	36	50	360	116	106
B	3/8"	48	50	5/16	**	36	45	94	86	88
B		48	60	**	1/2	36	45	94	86	88
B		72	50	5/16	**	36	45	116	86	90
B		72	60	**	1/2	36	45	116	86	90
B		96	50	5/16	**	36	47	142	86	90
B		120	50	5/16	**	36	48	178	86	94
B		144	50	5/16	**	36	48	212	88	94
B		168	40	3/8	**	36	48	233	100	108
B		192	40	3/8	**	36	50	248	104	110
B		240	30	3/8	**	48	60	264	106	110
B	1/2"	48	40	3/8	**	36	48	110	90	90
B		48	50	**	1/2	36	48	110	90	90
B		72	40	3/8	**	36	48	130	98	94
B		72	50	**	1/2	36	48	130	98	94
B		96	40	3/8	**	36	48	162	98	108
B		120	40	3/8	**	36	48	185	98	108
B		144	40	3/8	**	36	48	210	100	108
B		168	30	3/8	**	36	54	240	104	112
B		192	30	3/8	**	48	54	262	106	112
C		240	20	3/8	**	48	54	314	118	112

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TABLE II. (Cont'd)

Type II. Gap Frame Machines

Class A - * Gap Depth Not Less Than 6 Inches
Class B - * Gap Depth Not Less Than 18 Inches
Class C - * Gap Depth Not Less Than 24 Inches
Class D - * Gap Depth Not Less Than 36 Inches

Style 1 - Low Rake Angle
Style 2 - High Rake Angle

Machine Class	Machine Size Shearing Capacity, (Inches) Not Less Than: Workpiece Thickness	Workpiece Width	Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than: Style 1	Style 2	Back Gage Capacity, (Inches) Not Less Than	Front Gage Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics, (Inches) Not More Than: Width	Depth (w/o Front Ga.)	Height
B,C	5/8"	48	40	17/32	**	36	48	112	94	100
B,C		72	40	17/32	**	36	48	137	97	108
B,C		96	40	17/32	**	36	48	164	97	108
B,C		120	30	17/32	**	36	54	190	106	108
B,C		144	30	17/32	**	36	54	214	106	108
C		192	20	17/32	**	48	54	266	116	108
B,C	3/4"	72	30	3/8	1/2	36	60	144	106	112
B,C		96	30	3/8	1/2	36	60	168	106	112
B,C		120	30	3/8	1/2	36	60	192	106	112
B,C		144	30	3/8	1/2	36	60	216	106	112
C		168	20	1/2	**	36	60	252	116	134
C		192	20	7/16	**	36	60	276	123	134
D		240	15	1/2	**	36	60	332	148	144
C	1"	48	20	1/2	**	36	60	125	116	112
B		48	30	**	5/8	36	60	125	116	112
C		72	20	1/2	**	36	60	146	116	112
B,C		72	30	**	5/8	36	60	146	116	112
C		96	20	1/2	5/8	36	60	180	116	134
C		120	20	1/2	5/8	36	60	204	122	134
C		144	20	1/2	5/8	48	60	228	122	144
C		168	20	1/2	5/8	48	60	260	123	158
D		192	15	5/8	**	48	60	284	126	158
C	1-1/4"	48	20	3/4	**	36	60	125	116	112
C		72	20	3/4	**	36	60	172	116	112
C		96	20	17/32	3/4	36	60	190	146	144
C,D		120	15	9/16	**	48	60	216	146	158
C,D		144	15	9/16	**	48	60	238	148	164
D		168	15	11/16	**	48	60	260	148	164

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TABLE II.(Cont'd)

Type II. Gap Frame Machines

Class A - * Gap Depth Not Less Than 6 Inches
Class B - * Gap Depth Not Less Than 18 Inches
Class C - * Gap Depth Not Less Than 24 Inches
Class D - * Gap Depth Not Less Than 36 Inches

Style 1 - Low Rake Angle
Style 2 - High Rake Angle

Machine Class	Machine Size Shearing Capacity, (Inches) Not Less Than		Shearing Speed, Ram Strokes Per Minute, Not Less Than	Rake (Shear) Angle, Upper Shear Knife (In. Per Ft.) Not More Than		Back Gage Capacity, (Inches Not Less Than	Front Gage Capacity, (Inches) Not Less Than	Machine Dimensional Characteristics, (Inches) Not More Than		
	Workpiece Thickness	Workpiece Width		Style 1	Style 2			Width	Depth (w/o Front Ge.)	Height
C,D	1-1/2"	72	15	5/8	**	48	60	164	148	164
C,D		96	15	3/4	**	48	60	190	148	164
C,D		120	15	3/4	**	48	60	212	148	166
C,D		144	15	3/4	**	48	60	238	148	166

* The gap depth of the frame is measured from the throat of the gap to the face of the stationary shear knife.

** No machine available in this style.

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Navy - SH
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Review Activities:

Army - WC
Navy - SH, YD
Air Force - 84

User Activities:

Army - AV, ME, SM
Navy - OS

Preparing Activity:

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CIVIL AGENCY INTEREST:

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